Roll No. .....

TEE–401

B. TECH. (EE) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

ELECTRICAL MACHINES—II

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Explain torque-slip characteristic of three-phase induction motor. (CO4)

OR

(b) Explain torque-slip characteristic of three-phase induction motor. (CO4)

2. (a) Discuss condition for maximum starting and running torque for three-phase induction motor. (CO3)

OR

(b) Describe analogy of induction motor with transformer in details. (CO3)

3. (a) State starting methods of three-phase induction motor and explain any *one* of them in details. (CO1)

P. T. O.

OR

(b) A 500 V, 6 pole, 50 Hz, three-phase induction motor develops 20 kW. When running at 995 r.p.m. the p.f. is 0.87 lagging. Calculate (i) slip (ii) rotor copper loss (iii) the total input if stator loss is 1500 W (iv) line current (v) the rotor current frequency. (CO1)

4. (a) Discuss in details no-load and blocked rotor test of three-phase induction motor. (CO3)

OR

(b) A 3-phase induction motor has a starting torque of 150% of full-load and maximum torque of 250 % of full-load torque. Neglect stator resistance and assume constant rotor resistance. Compute (i) Slip at maximum torque (ii) Full load slip. (CO3)

5. (a) Draw the equivalent circuit of induction motor and obtain the equation for output power. (CO2)

OR

(b) The efficiency of a 400 V, 3-phase, 6 pole induction motor draws a line current of 80 A at 0.75 p.f. at 4% slip is 85%. Calculate shaft output and shaft torque. (CO2)

Roll No. .....

TMA–401

B. TECH. (ELECTRICAL ENGINEERING)

(FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

MATHEMATICS—III

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) An urn contains 10 white, 15 red and 25 black marbles. Determine the probability of drawing (i) Either a white or a red marble. (ii) Either a white or a red or a black marble. (CO1)

OR

(b) A bag contains four black and five red balls. If three balls from the bag are chosen at random, what is the chance that they are all black ? (CO1)

2. (a) Suppose that X is a random variable with $E(x) = 10$ and var $(x) = 25$. Find the positive number a and b such that $Y = aX - b$ has mean 0 and variance 1. (CO1)

P. T. O.

OR

(b) The mean and variance of a binomial variate X with parameters n and p are 16 and 8.

Find the following : (CO1)

(i) $P(X = 0)$

(ii) $P(X = 1)$

(iii) $P(X \geq 2)$

3. (a) The diameter of an water pipe, say X, is assumed to be a continuous random variable with probability density function $f(x) = 6x(1 - x),\ 0 \leq x \leq 1$: (CO2)

(i) Check validity of $f(x)$ to be p.d.f.

(ii) Determine a number b such that $P(X < b) = P(X > b)$

OR

(b) If the M. G. F. of a random variable X is of the form $M_X(t) = (0.4\,e^t + 0.6)^8$. What is the M. G. F. of $Y = 3x + 2$? Also find E[X]. (CO2)

4. (a) The life in hours of a certain radio tube has the probability density function given by : (CO2)

$$f(x) = \begin{cases} \dfrac{100}{x^2}, & \text{for } x \geq 100 \\ 0, & \text{for } x < 100 \end{cases}$$

Find :

(i) The distribution function $F(x)$.

(ii) The probability that none of three such tubes in a given radio tube will have to be replaced during the first 150 hours.

OR

(b) The number of airplane landings at an airport in a minute (X) and their probabilities are given by : (CO2)

X_i	$p(x_i)$
0	0.02
1	0.15
2	0.22
3	0.26
4	0.17
5	0.14
6	0.04

Find the mean and standard deviation of X.

5. (a) If X is random variable for probability mass function is : (CO1)

$$p(x) = \begin{cases} \frac{x}{15}, & x = 1, 2, 3, \ldots.. \\ 0, & \text{elsewhere} \end{cases}$$

find (i) P(X = 1 or 2) and (ii) $P\left(\frac{\frac{1}{2} < X < \frac{5}{2}}{X > 1}\right)$.

OR

(b) A random variable X has the following probability function values :

(CO1)

X	$P(X = x)$
0	0
1	b
2	$2b$
3	$2b$
4	$3b$
5	b^2
6	$2b^2$
7	$7b^2 + b$

(i) Find the value of b.

(ii) Determine the distribution function of r.v. X.

(iii) Evaluate $P(X < 6), P(X \geq 6)$ and $P(0 < X < 5)$.

Roll No. .....

TEE–402

B. TECH. (EE) (FOURTH SEMESTER)

MID SEMESTER EXAMINATION, April, 2023

INTRODUCTION TO ELECTRICAL ENERGY SOURCES

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any ***one*** of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) What are Conventional and Non-conventional energy sources ? (CO1)

OR

(b) Differentiate between peak load and base load power plant. (CO1)

2. (a) Discuss why the overall efficiency of a thermal plant is very low. (CO2)

OR

(b) Explain Flue gas cycle flow diagram of a modern coal based thermal plant. (CO2)

3. (a) With the help of a neat diagram, explain an Electrostatic Precipitator. (CO2)

OR

(b) Discuss why more plants use pulverized coal. (CO2)

P. T. O.

4. (a) What materials are suitable as moderator materials in a reactor ? What are the advantages of nuclear power plant over conventional steam plant ?

(CO2 and CO3)

OR

(b) Define binding energy and mass defect. (CO2 and CO3)

5. (a) With the help of neat diagram, explain the working of PWR. (CO3)

OR

(b) A U^{235} nucleus is bombarded by a neutron, resulting in its fission into Barium 137 and Krypton 97 nuclei. Write the complete nuclear equation and find the amount of energy liberated in the reaction. (CO3)

Roll No. .....

TEE–403

B. TECH. (EE) (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

SIGNALS AND SYSTEMS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) What do you mean by signals and systems ? Discuss linear and time invariant systems. (CO1)

OR

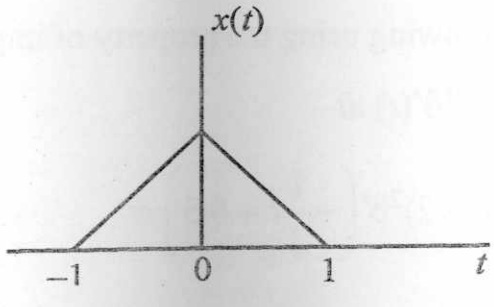
(b) Find the fundamental period of the following signals : (CO1)

(i) $x(n) = \cos(0.1\pi n)$

(ii) $x(n) = e^{j6\pi n}$

(iii) $x(n) = e^{j\frac{2\pi}{3}n} + e^{j\frac{3\pi}{4}n}$

2. (a) For the triangular signal, $x(t)$ as shown below, find (i) $x(2t + 2)$ (ii) $x(t - 3)$: (CO1)



P. T. O.

OR

(b) If energy of signal $x(t)$ is E, then find the energy of the following signals : (CO1)

(i) $x(-t)$

(ii) $x(t - T)$

3. (a) Check whether the following systems are linear or not : (CO2)

(i) $y(t) = tx(t)$

(ii) $y(t) = x^2(t)$

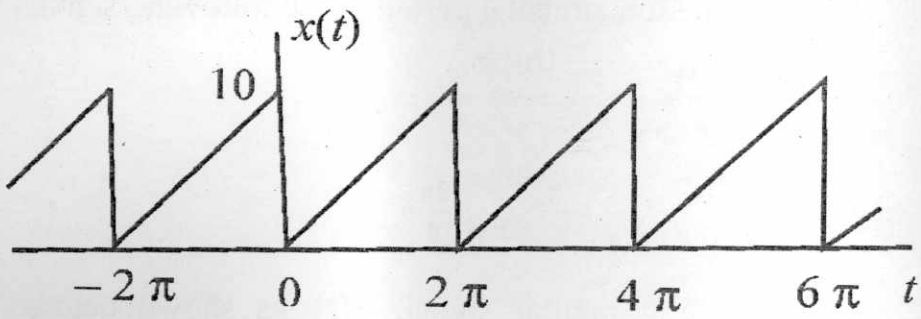
OR

(b) Check whether the following systems are time invariant or not : (CO2)

(i) $y(t) = \sin[x(t)]$

(ii) $y(t) = x(2t)$

4. (a) Find Fourier series expansion of the waveform shown below : (CO3)



OR

(b) Find the following using the property of impulse signals : (CO3)

(i) $\int_{-\infty}^{+\infty} e^{-2t}\delta'(t)\, dt$

(ii) $\int_{-\infty}^{+\infty} (t-2)^2\delta'\left(-\frac{1}{3}t + 0.5\right) dt$

5. (a) If $x(t) * h(t) = y(t)$; then show that : (CO3)

$$x(t) * h(t - t_0) = y(t - t_0) = x(t - t_0) * h(t).$$

OR

(b) If $x(t) * h(t) = y(t)$, then show that : (CO3)

$$\frac{d[x(t)]}{dt} * h(t) = x(t) * \frac{d[h(t)]}{dt} = \frac{d[y(t)]}{dt}.$$

50

Roll No. .....

TEE–404

B. TECH. (EE) (FOURTH SEMESTER)
MID SEMESTER EXAMINATION, April, 2023

DIGITAL ELECTRONICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) Convert methodically the following numbers to decimal numbers :

(i) $(F8E6)_{hex}$

(ii) $(258)_{Octal}$

(CO1)

OR

(b) Write the gray codes of the following decimal numbers : (CO1)

(i) $(0)_{10}$

(ii) $(8)_{10}$

(iii) $(10)_{10}$

(iv) $(15)_{10}$

P. T. O.

2. (a) Why is the NAND gate called a universal building block ? Realize the following logic operation using NAND gates : (CO2)

(i) NOT,

(ii) AND,

(iii) OR,

(iv) EXOR

OR

(b) Is NOR gate a universal building block ? Realize the following logic operations using NOR gates : (CO1)

(i) NOT,

(ii) OR,

(iii) AND,

(iv) EXOR

3. (a) Given five variables A through E, what is the fundamental product for each of these input conditions : (CO2)

(i) A B C D E = 0 0 1 1 0

(ii) A B C D E = 0 1 1 0 1

(iii) A B C D E = 1 0 0 0 1

OR

(b) Convert Binary 1 1 0.0 0 1 to a decimal number. (CO2)

4. (a) Describe the De Morgan's theorems. Prove any *one* of the two theorems. (CO2)

OR

(b) Write the basic Boolean laws and their respective Boolean expressions. (CO2)

5. (a) Minimize the four variable logic function using Karnaugh map :

$$f(w, x, y, z) = \Sigma m\,(0, 1, 2, 3, 7, 8, 9, 10, 11, 12, 13)$$

(CO3)

OR

(b) Minimize the logic function : (CO3)

$$f(A, B, C, D) = \Pi M\,(4, 6, 10, 12, 13, 15) \text{ in POS form.}$$